WORKED SOLUTION

STEP III 2001, Question 9

Mechanics • Friction and SHM

STEP III 2001, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q9

The key idea

The beams are d apart along a line inclined at α to the horizontal, so the plank lies at angle α. As the plank slides, its weight shifts from B_1 to B_2, and the two normal reactions change *linearly* with the position of the centre of gravity. Since the two coefficients of friction differ, the total friction changes linearly too, and a restoring force proportional to displacement means simple harmonic motion. The condition $\mu_1 + \mu_2 = 2\tan\alpha$ is exactly what puts the centre of that motion halfway between the beams.

Solution

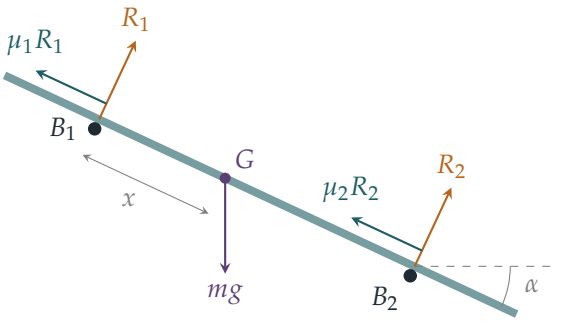
The model

Treat the plank as a rigid rod of mass m that stays in contact with both beams, which we treat as points. The line from B_1 to B_2 has length

$$\sqrt{d^2\sin^2\alpha + d^2\cos^2\alpha} = d$$

and makes angle α with the horizontal, so the plank lies along that line, sloping down from B_1 to B_2. Let x be the distance of the centre of gravity G from B_1, measured along the plank down the slope, so that $x = 0$ at release and G is over B_2 when $x = d$.

While $0 \le x \le d$, G lies between the beams. The forces are the weight mg at G; normal reactions R_1, R_2 perpendicular to the plank at B_1, B_2; and, while the plank moves down the slope, friction $\mu_1 R_1$ and $\mu_2 R_2$ up the slope.



The normal reactions

The plank does not move perpendicular to itself, so resolving perpendicular to the plank,

$$R_1 + R_2 = mg\cos\alpha.$$

It does not rotate either, so taking moments about G (the weight and both friction forces act along lines through G or along the plank, so they have no moment),

$$R_1 x = R_2(d - x).$$

Solving these two equations,

$$R_1 = \frac{mg\cos\alpha\,(d-x)}{d}, \qquad R_2 = \frac{mg\cos\alpha\,x}{d}.$$

Both are non-negative for $0 \le x \le d$, so the plank does stay on both beams.

The equation of motion

Resolving down the slope, while the plank moves down it,

$$\begin{aligned} m\ddot{x} &= mg\sin\alpha - \mu_1 R_1 - \mu_2 R_2 \\ &= mg\cos\alpha\left[\tan\alpha - \mu_1 - \frac{(\mu_2-\mu_1)x}{d}\right]. \end{aligned}$$

Now use the given condition: $\mu_1 + \mu_2 = 2\tan\alpha$ gives $\tan\alpha - \mu_1 = \frac{1}{2}(\mu_2 - \mu_1)$. Hence

$$\ddot{x} = -\omega^2\left(x - \tfrac{1}{2}d\right), \qquad \text{where} \qquad \omega^2 = \frac{g(\mu_2-\mu_1)\cos\alpha}{d} > 0,$$

the inequality holding because $\mu_2 > \mu_1$ and $\cos\alpha > 0$.

Ask yourself Does the plank actually start to move, and is the friction always pointing up the slope?

At $x = 0$, $\ddot{x} = \frac{1}{2}\omega^2 d > 0$, so the plank does start to slide down. The equation above is simple harmonic motion about $x = \frac{1}{2}d$, and with $x = 0$, $\dot{x} = 0$ at $t = 0$ its solution is

$$x = \tfrac{1}{2}d\,(1 - \cos\omega t), \qquad \dot{x} = \tfrac{1}{2}d\,\omega\sin\omega t.$$

For $0 < t < \pi/\omega$ we have $\dot{x} > 0$, so the plank really is moving down the slope throughout, and friction acting up the slope was the right assumption. Also $0 \le x \le d$ throughout, so G stays between the beams and the expressions for R_1, R_2 remain valid.

Coming to rest

The velocity is first zero again when $\omega t = \pi$, and then $x = \frac{1}{2}d(1 - \cos\pi) = d$: the centre of gravity is directly over B_2. At that instant $R_1 = 0$ and $R_2 = mg\cos\alpha$, so the plank rests on B_2 alone. It stays there, because friction at B_2 can supply up to $\mu_2 mg\cos\alpha$, while the component of weight down the slope is $mg\sin\alpha$, and

$$\mu_2 = \tfrac{1}{2}(\mu_1 + \mu_2) + \tfrac{1}{2}(\mu_2 - \mu_1) > \tfrac{1}{2}(\mu_1 + \mu_2) = \tan\alpha.$$

So the plank comes to rest with its centre of gravity over B_2 after time

$$\frac{\pi}{\omega} = \pi\left(\frac{d}{g(\mu_2 - \mu_1)\cos\alpha}\right)^{\frac{1}{2}},$$

as required.

Checks

- If $\mu_2 - \mu_1$ is small, the plank barely decelerates and the time is long, as the formula says. Each μ is then close to $\tan\alpha$, so the net force is close to zero everywhere.
- We stepped the equation of motion numerically for random α, d, μ_1 (with $\mu_2 = 2\tan\alpha - \mu_1$), stopping when the velocity first returned to zero. For example, with $\alpha = 0.7$, $d = 2$, $\mu_1 = 0.3$ and $g = 9.8$, the predicted time is about 1.56 seconds. In every case the stopping time and position agreed with π/ω and $x = d$ to within 10^{-5} (relative).

Common slips

- Taking $R_1 = R_2 = \frac{1}{2}mg\cos\alpha$. The reactions depend on where G is, and that dependence is what produces the oscillation.
- Stopping at "the velocity is zero" without checking that the plank then stays at rest: you need $\mu_2 \ge \tan\alpha$.

Beyond STEP

The motion is half a cycle of an oscillator whose "spring" comes from friction varying with load. Systems in which friction depends on position or velocity lead to rich behaviour, including the stick–slip oscillations behind squeaking hinges and bowed violin strings. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction, rigid-body equilibrium and simple harmonic motion with care.

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